Bounded Residual Quaternion Attitude Control with Non-Spiking Conductance-Based Neurons and Online Decoder Learning

Gandhi Contreras*, Steven Sandoval*, Luis Garcia†
*New Mexico State University (NMSU), Las Cruces, NM, USA
†U.S. Air Force, Los Alamos, NM, USA

***Abstract*—We study a compact neuromorphic residual architecture for quaternion attitude tracking under bounded actuator commands. A fixed quaternion proportional–derivative (PD) controller is augmented by six paired continuous coupled membrane–gate (CMG) units: a conductance-based membrane coupled to a gating variable, in the graded, non-spiking regime that analog neuromorphic circuits realize directly. Their gate-derived activities are decoded through eighteen weights adapted online by prescribed error sensitivity (PES), and the decoded residual is *added* to the saturated baseline, so the baseline is recovered exactly when the residual authority is zero. We separate three mechanisms that are usually conflated — intrinsic conductance modulation, decoder learning, and output authority — and establish bounded neural states, bounded command deviation, projected decoder boundedness, and a restricted frozen-feature learning result. In ten matched 45-s MuJoCo cases spanning five missions, a Skydio X2 model and a Crazyflie 2.0 model completed with an explicit four-rotor actuation interface track 17.3–60.6% more accurately than PD, with zero actuator-bound occupancy outside brief Hardstep intervals. Direct and Nengo realizations agree to 2.013×10^{-6} in relative mean error. Controlled ablation attributes the improvement to online decoder adaptation rather than to conductance modulation in the tested gate-gain regime — a result that constrains, rather than confirms, the role of the neuron model. Rerunning every cell under measurement noise leaves the seed standard deviation of the reduction below 0.031% and the range at 18.1–55.8%, and a hold-equilibrium study under intermittent disturbance shows the decoder accumulating a correction that is reused across episodes. The study is in simulation and establishes boundedness rather than closed-loop stability.**

I. Introduction

Learning-based augmentation can retain useful classical feedback structure while adapting a correction that is difficult to specify a priori. Residual robot control is established [1]; quadrotor examples include geometric $\mathcal{L}_1$ augmentation with performance guarantees [2] and PID-plus-PPO attitude residuals [3]. Thus, neither additive residual learning nor online flight adaptation is novel in isolation.

Neuromorphic control offers a complementary implementation approach. Nengo supports the Neural Engineering Framework (NEF) and online decoder learning [4]. PES-based adaptive control has been demonstrated on manipulators [5], [6], and a Nengo spiking PID controller with PES-based self-tuning has been validated on a free-flying UAV [7]. Building on that line, we investigate a specific continuous membrane–gate basis for a bounded quaternion-torque residual, with a fixed classical controller and adaptation confined to its decoder.

The central question is whether this small stateful basis can improve tracking, and which of its components explains the improvement. The paper contributes: (i) a code-aligned CMG–PES architecture with paired currents, explicit state and readout equations, and a fully specified actuator interface (Fig. 1); (ii) boundedness and exact baseline-recovery properties, stated at the strength the analysis supports; (iii) a matched five-mission, two-model study in which the direct and Nengo realizations agree to 2×10^{-6}; and (iv) a controlled ablation that separates conductance modulation, decoder learning, and output authority, and identifies which of the three carries the effect. The third and fourth are the substantive contributions: the architecture is small and its components are individually replaceable, so establishing *which* part does the work is more useful than reporting that the whole improves tracking.

The CMG unit is conductance-based: a leaky membrane whose effective conductance is modulated by a gating variable with its own first-order dynamics and threshold self-coupling. Structurally this is the Hodgkin–Huxley gating lineage rather than the leaky integrate-and-fire one, operated in a graded, non-spiking regime; there is no firing threshold and no reset event. That regime is not a weakening of the neuromorphic premise but the one in which neuromorphic engineering began: analog silicon neurons implement membrane leak, synaptic conductance, and sigmoidal gating directly as device physics [8], without the comparator and event-routing hardware that spiking requires. We therefore use "neuromorphic" in that conductance-based sense throughout. No power or device-level claim is made here; the implementation is a digital simulation, and Sec. V states what would be required to test the analog case. PES is an established rule [9]; what is new here is the basis it decodes and the controlled separation of the mechanisms that basis contributes.

II. Simulation Models, Attitude Baseline, and Actuator Interface

A. MuJoCo Platforms and CF2 Actuation Extension

We use Crazyflie 2.0 (CF2) and Skydio X2 (X2) models derived from MuJoCo Menagerie [10]. Each vehicle has six rigid-body degrees of freedom: three translations and three rotations, represented by seven free-joint position coordinates

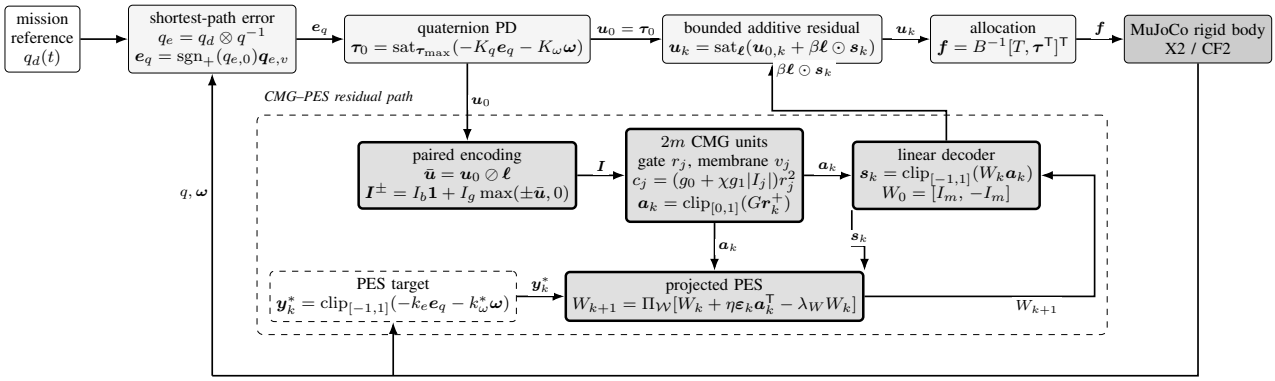


Fig. 1. CMG–PES residual architecture. The light blocks are the fixed classical path: error (6), PD law (12), and allocation (14) are never adapted. The heavy blocks encode the clipped baseline as paired currents (16), evolve $2m$ membrane–gate units (19), and decode a bounded residual (23) that is *added* to the saturated baseline (27). Only W_k adapts, through (26); $\beta = 0$ recovers the baseline exactly.



Fig. 2. Vehicles and simulation models. Left: CF2 hardware and MuJoCo model. Centre: X2 hardware and MuJoCo model. Right: both geometries at relative scale. The axis triads are body-frame axes: they are rigidly attached to the vehicle and therefore rotate with it relative to the inertial (world) frame (cf. Q: body $\to$ inertial in Sec. II).

(position and unit quaternion) and six velocity coordinates. The attitude controller requests three body torques; a separate classical height loop requests collective thrust. Thus attitude tracking is evaluated in a freely translating model, not an attitude-only joint model.

The upstream CF2 interface provides collective thrust and three abstract body-moment channels; its documentation identifies the original actuator limits as arbitrary and requiring tuning [11]. Our four-rotor actuation extension replaces that interface with four nonnegative, spatially located thrust inputs while retaining the source mass, inertia, and visual geometry. Rotor moment arms and signed reaction-torque coefficients define an explicit allocation matrix, making the coupling between collective thrust, attitude torque, and individual actuator limits testable. This is a model-interface contribution, not a complete aerodynamic identification of CF2.

The upstream CF2 model ships without a rotor-level actuation interface, and its 0.35 N total thrust limit falls short of the 0.568 N measured on stock hardware at 93.75% PWM [12]. We therefore complete the model with an explicit four-rotor interface and set the per-rotor limit so the total is 0.525 N (Table II), which remains about 7.5% *below* that measured value. The correction is toward the manufacturer's own measurement, not beyond it: it raises modeled maximum thrust/weight from 1.321 to 1.982 at $g = 9.81$ m/s^2, a figure the physical vehicle exceeds. For physical perspective, constant-height flight at a single-axis 45° tilt requires $T/(Mg) = 1/\cos 45^\circ = \sqrt{2}$ even before reserving differential-thrust margin. The unscaled model cannot meet that idealized requirement; this does not establish such a limitation for stock CF2 hardware. The scale changes a simulation thrust bound, not rotor speed, stock propulsion capability, or tracking performance by 50%. Mass, inertia, and reaction coefficients are unchanged. PD and CMG use the same model and actuator bounds; any benefit from increased model authority is therefore distinct from the benefit of neural augmentation.

The masses agree with published specifications, but this does not validate all dynamic parameters (Table III). Bitcraze's historical measurement of 57.9 gram-force corresponds to approximately 0.568 N; the evaluated 0.525 N limit is about 7.5% below that operating point [12]. This supports a plausible thrust scale, not hardware validation, and does not establish that a stronger-than-stock motor is required. CF2 inertia is inherited from Menagerie, which cites system-identification work [11]; the rotor-extension moment arms and $|\kappa_i| = 0.005$ m remain design assumptions. X2 inertia, 13 N rotor limits, and $|\kappa_i| = 0.0201$ m are simulation parameters, not confirmed manufacturer characteristics.

Geometric dimensions also require like-for-like definitions. Bitcraze lists $92 \times 92 \times 29$ mm motor-to-motor/height dimensions [13]; our sites give an opposite-motor separation of 91.92 mm. Skydio lists an unfolded flying envelope of approximately $663 \times 569 \times 211$ mm [14], whereas the model's rotor centers span 280×360 mm. Center spacing and overall flying envelope are different quantities and do not by themselves establish geometric agreement or mismatch. Propulsion identification, motor and battery dynamics, and experimental validation remain necessary for hardware transfer; no endurance or

TABLE I

NOMENCLATURE. PARENTHESISED TAGS GIVE THE DEFINING EQUATION; VECTORS ARE BOLD; $\mathrm{clip}_{[a,b]}$ AND $\mathrm{sat}_{\boldsymbol{\ell}}$ ACT ELEMENTWISE.

Attitude, plant, and notation					
$q,\ q_d$	state / reference quaternion	q_e	error quaternion, $q_d \otimes q^{-1}$	$q_{e,0}, \boldsymbol{q}_{e,v}$	scalar and vector parts
$\boldsymbol{e}_q$	shortest-path error (6)	θ_q	error angle (6)	$\boldsymbol{\omega}$	body angular rate
J	inertia tensor	Q	body-to-inertial rotation	$\otimes$	quaternion product
$\boldsymbol{\omega}_d, \boldsymbol{\omega}_e$	reference and error rate (8)	$R(\cdot)$	rotation matrix of a quaternion	$z, z_d, \dot{z}$	height, reference, rate (15)
$\odot, \oslash$	elementwise $\times$, $\div$	sgn_+	sign, with $\mathrm{sgn}_+(0) = 1$	σ	logistic function
I_m	identity of order m	$\mathbb{R}$	real numbers	Π	Euclidean projection
Fixed baseline and actuator interface					
$\boldsymbol{\tau}_0 = \boldsymbol{u}_0$	baseline torque (12)	K_q, K_ω	PD gains (12)	$\boldsymbol{\tau}_{\max}$	per-axis torque bound
ω_n, ζ	design frequency, damping	$\boldsymbol{\ell}$	per-channel command bound	$\mathrm{sat}_{\boldsymbol{\ell}}$	saturation at $\pm\boldsymbol{\ell}$
B	allocation matrix (13)	$\boldsymbol{f}$	rotor forces (14)	T	collective thrust (15)
κ_i	rotor reaction coefficient	m	torque channels, $m = 3$	M	vehicle mass
$\boldsymbol{f}_{\max}$	per-rotor force bound	g	gravity, $9.81\,\mathrm{m/s^2}$	$\mathbf{1}$	vector of ones
Coupled membrane–gate unit					
v_j	membrane state (19)	r_j	gate state (19)	c_j	shunt conductance (19)
$\bar{\boldsymbol{u}}$	normalized baseline (16)	$\boldsymbol{I}^{\pm}, \boldsymbol{I}$	paired currents (16)	I_b, I_g	current offset and gain
τ, τ_r	membrane, gate time constants	v_{th}	gate threshold	$v_{\mathrm{rest}}, v_{\mathrm{reset}}$	rest level, reversal level
γ	gate self-coupling	α_h	logistic sharpness	χ	conductance switch, $\chi \in \{0, 1\}$
g_0, g_1	conductance gains (distinct from g)	G	activity gain, $G = 10^5$	h	integration step
$b_{j,k}$	gate target (21)	$r^+_{j,k}, v^+_{j,k}$	updated states (22)	$\lambda_{j,k}, d_{j,k}$	frozen membrane coefficients
Decoder, online learning, and residual authority					
$\boldsymbol{a}_k$	clipped activity (23)	W_k	decoder (23)	W_0	initial decoder, $[I_m, -I_m]$
$\boldsymbol{s}_k$	clipped readout (23)	$\boldsymbol{y}^*_k$	PES target (26)	$\boldsymbol{\varepsilon}_k$	PES error (26)
k_e, k^*_ω	target gains (26)	η	per-step learning rate	λ_W	per-step decoder decay
$\Pi_{\mathcal{W}}, W_{\max}$	projection and weight bound	β	residual authority (27)	$\boldsymbol{u}_k, \boldsymbol{\delta}_k$	applied command, deviation (27)
Reported metrics					
$\bar{\theta}_q$	mean quaternion-angle error	S	actuator-bound occupancy [%]	n	runs per reported cell

TABLE II

COMPILED SIMULATION-MODEL CHARACTERISTICS. CF2 AUTHORITY IS SCALED RELATIVE TO THE UPSTREAM MODEL LIMIT, NOT A MANUFACTURER RATING. HARDWARE COMPARISONS ARE GIVEN IN TABLE III.

Property	CF2 rotor model	X2 model
Mass M (kg)	0.027	1.325
Rotor $\lvert p_x\rvert, \lvert p_y\rvert$ (m)	$0.0325, 0.0325$	$0.14, 0.18$
Reaction magnitude $\lvert\kappa_i\rvert$ (m)	0.005	0.0201
Unscaled rotor limit (N)	0.0875	13
Main-test authority scale	1.5	1
Main-test rotor limit (N)	0.13125	13
Main-test total limit (N)	0.525	52
Maximum thrust/weight	$1.321 \to 1.982$	4.001
Rigid-body integrator	RK4	Euler
Experiment timestep (s)	0.0005	0.0005
Motor activation/RPM states	None	None

TABLE III

SIMULATION PARAMETERS VERSUS MANUFACTURER INFORMATION. "NOT REPORTED" REFERS TO THE MANUFACTURER SOURCES REVIEWED, NOT PROOF THAT THE PARAMETER IS UNAVAILABLE ELSEWHERE.

Quantity	Simulation	Manufacturer information
CF2 mass	27 g	27 g [13]
X2 mass	1325 g	1325 g with battery [14]
CF2 total thrust	$0.350 \to 0.525$ N	0.568 N measured at 93.75% PWM[a] [12]
X2 rotor limit	13 N per rotor	Not reported [14]
CF2 motor response	No motor dynamics	~ 180 ms: $0 \to 25{,}000$ RPM[b] [15]
Inertia tensors	CF2 inherited; X2 geometry-derived	Not reported in product specifications [13], [14]

[a]Historical total-thrust measurement with a power-supply test setup, not a guaranteed continuous battery-powered rating. [b]One motor's reported full-command response with a charged battery; not an identified first-order time constant.

electrical-power fidelity is claimed.

MuJoCo [16] integrates rigid-body motion, gravity, site-applied actuator wrenches, and enabled passive/contact forces. Both models specify air density $1.225\ \mathrm{kg/m^3}$ and viscosity 1.8×10^{-5} Pa s; their default inertia-based fluid model [17] is a coarse drag approximation, not rotor-resolved aerodynamics. The runner overrides the timestep but retains each model's integrator. Actuators apply algebraic thrust commands without motor activation dynamics. Although gyro, accelerometer, and orientation sensors are defined, the evaluated feedback is read directly from `qpos`/`qvel`; optional measurement perturbations are separate runner settings. These experiments do not establish performance with a hardware state estimator.

B. Quaternion Convention and Error

Let $Q \in \mathbb{S}^3$ be the scalar-first Hamilton quaternion mapping body coordinates to inertial coordinates, and let $\boldsymbol{\omega}$ be body angular velocity. With physical inertia $\boldsymbol{J} \succ 0$ about the center of mass, expressed in body axes, the free-flight rotational balance is

$$\dot{Q} = \tfrac{1}{2} Q \otimes [0, \boldsymbol{\omega}^{\mathsf{T}}]^{\mathsf{T}}, \tag{1}$$

$$\boldsymbol{J}\dot{\boldsymbol{\omega}} = \boldsymbol{\tau}_{\mathrm{act}} + \boldsymbol{\tau}_{\mathrm{p}} + \boldsymbol{\tau}_{\mathrm{d}} - \boldsymbol{\omega} \times (\boldsymbol{J}\boldsymbol{\omega}). \tag{2}$$

Here $\boldsymbol{\tau}_{\mathrm{p}}$ denotes modeled passive torque and $\boldsymbol{\tau}_{\mathrm{d}}$ any applied disturbance; the nominal ideal rigid-body model sets both to zero. Contact, if present, contributes an additional external wrench. The requested torque and actuator torque are distinct signals in Fig. 2.

The controller conjugates MuJoCo's orientation and uses the passive quaternion $q = Q^{-1}$, so

$$\dot{q} = -\tfrac{1}{2}[0, \boldsymbol{\omega}^{\mathsf{T}}]^{\mathsf{T}} \otimes q. \tag{3}$$

The desired q_d uses the same convention. Define

$$q_e = q_d \otimes q^{-1} = [q_{e,0}, \boldsymbol{q}_{e,v}^{\mathsf{T}}]^{\mathsf{T}}, \tag{4}$$

$$\boldsymbol{e}_q = \operatorname{sgn}_+(q_{e,0})\boldsymbol{q}_{e,v}, \qquad \operatorname{sgn}_+(0) = 1, \tag{5}$$

$$\theta_q = 2\,\operatorname{atan2}(\|\boldsymbol{q}_{e,v}\|_2, |q_{e,0}|) \in [0, \pi]. \tag{6}$$

The reported error θ_q is the shortest rotation angle, invariant under $q_e \mapsto -q_e$. Sign selection introduces a discontinuity at $q_{e,0} = 0$; it does not establish robust hybrid global stability [18]. For a desired-frame rate $\boldsymbol{\omega}_d$ satisfying the corresponding passive kinematics, differentiation gives

$$\dot{q}_e = \tfrac{1}{2} q_e \otimes [0, \boldsymbol{\omega}_e^{\mathsf{T}}]^{\mathsf{T}}, \tag{7}$$

$$\boldsymbol{\omega}_e = \boldsymbol{\omega} - R(q_e)^{\mathsf{T}}\boldsymbol{\omega}_d, \qquad \dot{q}_{e,0} = -\tfrac{1}{2}\boldsymbol{q}_{e,v}^{\mathsf{T}}\boldsymbol{\omega}_e. \tag{8}$$

This is a kinematic identity, not a convergence result. Reference Euler triples parameterize q_d in the passive convention, rather than active body-to-world Euler commands.

C. Physical Inertia and Implemented PD Gains

MuJoCo stores principal moments $\boldsymbol{j}_{\mathrm{p}}$ in its inertial frame [19]. If R_{BI} maps that frame into body axes, the physical inertia is

$$\boldsymbol{J} = R_{BI}\,\operatorname{diag}(\boldsymbol{j}_{\mathrm{p}})R_{BI}^{\mathsf{T}}. \tag{9}$$

The evaluated implementation uses a tuning vector $\boldsymbol{j}_{\mathrm{tune}} = \boldsymbol{j}_{\mathrm{p}}$ directly, in the returned array order:

$$\boldsymbol{j}_{\mathrm{tune}}^{\mathrm{CF2}} = (2.3951, 2.3951, 3.2347)^{\mathsf{T}}10^{-5},$$
$$\boldsymbol{j}_{\mathrm{tune}}^{\mathrm{X2}} = (0.0607113, 0.0364684, 0.0254117)^{\mathsf{T}},$$

in $\mathrm{kg\,m^2}$. CF2's inertial axes coincide with its body axes. For X2 they do not; its physical body-frame tensor is approximately

$$\boldsymbol{J}_{\mathrm{X2}} = \begin{bmatrix} 0.0366517 & 0 & -0.0021000 \\ 0 & 0.0254117 & 0 \\ -0.0021000 & 0 & 0.0605280 \end{bmatrix} \ \mathrm{kg\,m^2}. \tag{10}$$

Thus the tuning vector must not be identified with $\operatorname{diag}(\boldsymbol{J})$ for both vehicles.

The fixed quaternion PD baseline is

$$\boldsymbol{k}_q = 2\boldsymbol{j}_{\mathrm{tune}} \odot \boldsymbol{\omega}_n^2, \qquad \boldsymbol{k}_\omega = 2\zeta\boldsymbol{j}_{\mathrm{tune}} \odot \boldsymbol{\omega}_n, \tag{11}$$

$$\boldsymbol{u}_0 = \operatorname{sat}_{\boldsymbol{\ell}}(-\boldsymbol{k}_q \odot \boldsymbol{e}_q - \boldsymbol{k}_\omega \odot \boldsymbol{\omega}), \tag{12}$$

where $\odot$ denotes componentwise multiplication, $\boldsymbol{\omega}_n = (4, 4, 3.5)^{\mathsf{T}}$ rad/s, $\zeta = 0.95$, and $\operatorname{sat}_{\boldsymbol{\ell}}$ clips component i to $[-\ell_i, \ell_i]$. For these rotor-interface models, the model-policy torque limits are $\ell_i = \max(j_{\mathrm{tune},i}a_i, 10^{-6}\ \mathrm{N\,m})$, with $\boldsymbol{a} = (12, 12, 10)^{\mathsf{T}}$ rad/s^2. These are controller torque bounds, not guaranteed feasible rotor torques. The law is quaternion feedback [20]; $\boldsymbol{\omega}_n$ parameterizes its gains but is not a verified set of closed-loop natural frequencies for coupled X2 dynamics. Desired-rate feedforward and gyroscopic cancellation are omitted. The identical fixed gains are retained for PD and its CMG augmentation.

For perspective only, take a constant reference, scalar proportional gain $\boldsymbol{k}_q = k_q\mathbf{1}$, the ideal plant ($\boldsymbol{\tau}_{\mathrm{p}} = \boldsymbol{\tau}_{\mathrm{d}} = 0$), no saturation or residual, and a trajectory confined to one smooth sign branch. Then $V = \frac{1}{2}\boldsymbol{\omega}^{\mathsf{T}}\boldsymbol{J}\boldsymbol{\omega} + 2k_q(1 - |q_{e,0}|)$ gives $\dot{V} = -\boldsymbol{\omega}^{\mathsf{T}}\operatorname{diag}(\boldsymbol{k}_\omega)\boldsymbol{\omega}$. This restricted identity does not prove stability of the anisotropic, saturated, moving-reference experiments.

D. Requested and Realized Wrench

For rotor positions $\boldsymbol{p}_i$ in body coordinates, parallel $+z$ thrust axes, and signed reaction coefficients κ_i, define

$$B = \begin{bmatrix} 1 & 1 & 1 & 1 \\ p_{1,y} & p_{2,y} & p_{3,y} & p_{4,y} \\ -p_{1,x} & -p_{2,x} & -p_{3,x} & -p_{4,x} \\ \kappa_1 & \kappa_2 & \kappa_3 & \kappa_4 \end{bmatrix}. \tag{13}$$

The signs follow $\boldsymbol{p}_i \times [0, 0, f_i]^{\mathsf{T}}$ and the actuator's local reaction torque $\kappa_i f_i$. Both models have zero body-frame center-of-mass x, y offsets, so the site's x, y coordinates also give the relevant center-of-mass moment arms. Figure 2 fixes the rotor ordering used by B; κ_i has units of meters, not angular velocity. The runner computes

$$\begin{aligned} \boldsymbol{f}_{\mathrm{req}} &= B^{-1}[T, \boldsymbol{u}^{\mathsf{T}}]^{\mathsf{T}}, \\ \boldsymbol{f}_{\mathrm{act}} &= \operatorname{clip}_{[\mathbf{0}, \boldsymbol{f}_{\max}]}(\boldsymbol{f}_{\mathrm{req}}), \\ [T_{\mathrm{act}}, \boldsymbol{\tau}_{\mathrm{act}}^{\mathsf{T}}]^{\mathsf{T}} &= B\boldsymbol{f}_{\mathrm{act}}. \end{aligned} \tag{14}$$

Here $\boldsymbol{u} = \boldsymbol{u}_0$ for PD; the CMG path in Section III modifies that requested torque before the same allocation and clipping. Each f_i is a commanded force in newtons, not a rotor-speed command. Exact wrench realization requires $B^{-1}[T, \boldsymbol{u}^{\mathsf{T}}]^{\mathsf{T}} \in [\mathbf{0}, \boldsymbol{f}_{\max}]$. A componentwise torque box is not generally contained in this collective-dependent feasible set. Clipped inverse allocation is used, not an optimization-based allocator [21].

Collective thrust remains classical, with $z_d = 1$ m:

$$T = \operatorname{clip}_{[0, 2.5Mg]}\left(\frac{M[g + 6(z_d - z) - 4\dot{z}]}{\max(R(Q)_{33}, 0.35)}\right). \tag{15}$$

This expression uses the model's $g = 9.81$ m/s^2. The height loop reads free-joint height and vertical velocity; the denominator compensates tilt with a lower guard. Its upper bound does not supersede individual rotor limits. The CMG path has no direct collective output, but can affect height through attitude, tilt compensation, and allocation saturation. The resulting shared plant interface isolates changes in requested attitude torque while preserving the same physical simulation for both controllers.

III. CMG State Dynamics and Online Decoder Adaptation

A. Encoding, States, and Numerical Update

Symbols are collected in Table I. For m torque channels and $2m$ units, normalize the clipped baseline and form positive/negative current pairs:

$$\begin{aligned} \bar{\boldsymbol{u}} &= \boldsymbol{u}_0 \oslash \boldsymbol{\ell}, \\ \boldsymbol{I}^{\pm} &= I_b\mathbf{1} + I_g \max(\pm\bar{\boldsymbol{u}}, \mathbf{0}), \\ \boldsymbol{I} &= [(\boldsymbol{I}^{+})^{\mathsf{T}}, (\boldsymbol{I}^{-})^{\mathsf{T}}]^{\mathsf{T}}. \end{aligned} \tag{16}$$

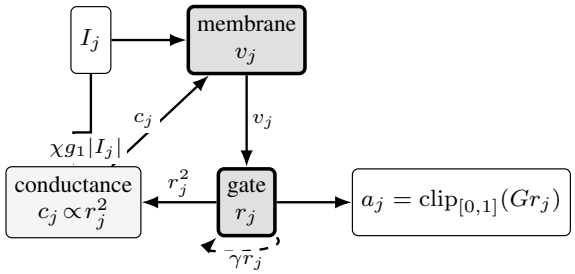


Fig. 3. Coupling topology of one CMG unit, cf. (19). The membrane sets the gate target; the gate acts back twice, shifting its own threshold through γ and shunting the membrane through $c_j \propto r_j^2$ ($\chi = 1$ makes that shunt input dependent). The dynamics are continuous: no spike, no reset, and only a_j leaves the unit.

Each unit has a dimensionless membrane state v_j and gate r_j:

$$\dot{r}_j = [\sigma(\alpha_h(v_j - v_{\mathrm{th}} + \gamma r_j)) - r_j]/\tau_r, \tag{17}$$

$$\dot{v}_j = -(v_j - v_{\mathrm{rest}})/\tau + I_j/\tau - c_j(v_j - v_{\mathrm{reset}}), \tag{18}$$

$$c_j = (g_0 + \chi g_1|I_j|)r_j^2, \qquad \chi \in \{0,1\}. \tag{19}$$

The logistic gate target depends on v_j, while r_j shifts that target and modulates membrane conductance (Fig. 3). Thus, both states participate even though the decoder reads the gate-derived activity. The model has no threshold-triggered reset or spike event; v_{reset} is a reversal level in a continuous conductance term. The gains g_0, g_1 have the units needed to make c_j a rate.

The actual integration is sequential, not exponential Euler for both states. From the old states compute

$$b_{j,k} = \sigma\Big(\mathrm{clip}_{[-60,60]}[\alpha_h(v_{j,k} - v_{\mathrm{th}} + \gamma r_{j,k})]\Big), \tag{20}$$

$$r_{j,k}^{+} = \mathrm{clip}_{[0,1]}[r_{j,k} + (h/\tau_r)(b_{j,k} - r_{j,k})]. \tag{21}$$

Then evaluate $c_{j,k} = (g_0 + \chi g_1|I_{j,k}|)(r_{j,k}^{+})^2$, $\lambda_{j,k} = \tau^{-1} + c_{j,k}$, and $d_{j,k} = (v_{\mathrm{rest}} + I_{j,k})/\tau + c_{j,k}v_{\mathrm{reset}}$. The frozen-coefficient membrane step is

$$v_{j,k}^{+} = e^{-\lambda_{j,k}h}v_{j,k} + (1 - e^{-\lambda_{j,k}h})d_{j,k}/\lambda_{j,k}. \tag{22}$$

This exactly solves the scalar membrane equation with those frozen coefficients; it does not establish unconditional stability of the coupled nonlinear closed loop. Assign $(v_{j,k+1}, r_{j,k+1}) = (v_{j,k}^{+}, r_{j,k}^{+})$.

B. Readout, PES, and Additive Residual

The within-step activity, initial decoder, and clipped readout are

$$\begin{aligned} \boldsymbol{a}_k &= \mathrm{clip}_{[0,1]}(G\boldsymbol{r}_k^{+}), \qquad W_0 = [I_m, -I_m], \\ \boldsymbol{s}_k &= \mathrm{clip}_{[-1,1]}(W_k\boldsymbol{a}_k). \end{aligned} \tag{23}$$

Pair subtraction retains directional information from a nonnegative activity basis; state memory does not imply instantaneous sign agreement after a command reversal. The teaching target and local update are

$$\boldsymbol{y}_k^{*} = \mathrm{clip}_{[-1,1]}(-k_e\boldsymbol{e}_{q,k} - k_\omega^{*}\boldsymbol{\omega}_k), \tag{24}$$

$$\boldsymbol{\varepsilon}_k = \boldsymbol{y}_k^{*} - \boldsymbol{s}_k, \tag{25}$$

$$W_{k+1} = \Pi_{\mathcal{W}}[W_k + \eta\boldsymbol{\varepsilon}_k\boldsymbol{a}_k^{\mathsf{T}} - \lambda_W W_k]. \tag{26}$$

Here $\mathcal{W} = [-W_{\max}, W_{\max}]^{m\times 2m}$, and η and λ_W are per-step coefficients, not continuous-time rates. Holding them fixed while changing h changes the learning dynamics. PES learns a prescribed residual target; it does not differentiate through the plant or directly minimize the closed-loop tracking cost. In particular, with a clipped readout this rule is not the exact gradient of the clipped instantaneous squared loss. It is a PES-style error-directed update with projection and leak.

Using W_k before the update, the requested command is

$$\boldsymbol{u}_k = \mathrm{sat}_{\boldsymbol{\ell}}(\boldsymbol{u}_{0,k} + \beta\boldsymbol{\ell}\odot\boldsymbol{s}_k), \quad \boldsymbol{\delta}_k = \boldsymbol{u}_k - \boldsymbol{u}_{0,k}, \qquad 0 \le \beta \le 1. \tag{27}$$

The canonical UAV case has $m = 3$, six CMG units, and eighteen decoder weights. Intrinsic conductance modulation (χ), synaptic/decoder adaptation (W_k), and output authority ($\beta\boldsymbol{\ell}$) are distinct.

C. Structural Properties and Their Scope

Proposition 1 (Bounded internal states)**.** *With the nonnegative conductance parameters in Table IV, $r_{j,0} \in [0,1]$, and $0 < h \le \tau_r$, the gate remains in $[0,1]$. The membrane iterate remains in the convex hull of its initial value, v_{reset}, and the interval $[v_{\mathrm{rest}} + I_b, v_{\mathrm{rest}} + I_b + I_g]$.*

Proof. The unclipped gate step is a convex combination of $r_{j,k}$ and $b_{j,k} \in [0,1]$. Since $I_j \in [I_b, I_b + I_g]$ and $c_j \ge 0$, d_j/λ_j is a convex combination of $v_{\mathrm{rest}} + I_j$ and v_{reset}. Equation (22) is a convex combination of $v_{j,k}$ and d_j/λ_j. Induction proves both statements. □

Proposition 2 (Bounded authority and baseline recovery)**.** *For $\boldsymbol{u}_{0,k} \in \mathcal{U}$ and any finite W_k, $\boldsymbol{u}_k \in \mathcal{U}$ and $|\delta_{k,i}| \le \beta\ell_i$. If $\beta = 0$, the requested baseline command is recovered at every step. If $W_k = 0$ at a particular step, it is recovered at that step, but learning need not keep $W_{k+1} = 0$.*

Proof. The outer clip enforces membership in $\mathcal{U}$. Its componentwise nonexpansiveness and $\mathrm{sat}_{\ell_i}(u_{0,k,i}) = u_{0,k,i}$ give $|\delta_{k,i}| \le |\beta\ell_i s_{k,i}| \le \beta\ell_i$. The recovery statements follow by substitution. □

The same allocator and state then give the same actuator output when $\beta = 0$. The proposition does not establish state safety, rotor headroom, stability, or a bound on tracking improvement. A bounded adversarial correction can still degrade or destabilize a nonlinear plant.

Proposition 3 (Decoder bound and restricted contraction)**.** *If $W_0 \in \mathcal{W}$, then* (26) *keeps $W_k \in \mathcal{W}$. If, additionally, the target and a nonzero activity $\boldsymbol{a}$ are fixed, leak is zero, and readout clipping and projection remain inactive, then*

$$\boldsymbol{\varepsilon}_{k+1} = (1 - \eta\|\boldsymbol{a}\|_2^2)\boldsymbol{\varepsilon}_k. \tag{28}$$

For a nonzero initial error, its norm contracts to zero when $0 < \eta\|\boldsymbol{a}\|_2^2 < 2$.

Proof. Projection gives the first result. Under the additional conditions, $(W_{k+1} - W_k)\boldsymbol{a} = \eta\boldsymbol{\varepsilon}_k\|\boldsymbol{a}\|_2^2$; subtracting this from the fixed target error gives (28). The scalar multiplier has modulus less than one under the stated condition. □

This is the constant-feature PES result [9], not a closed-loop convergence theorem. Since $\|\boldsymbol{a}\|^2 \le 2m$, $0 < \eta < 1/m$ is sufficient for that restricted model with nonzero activity; changing targets, loss of excitation, clipping, and leakage invalidate a direct application to the experiment. No unique identification of all decoder weights follows from one fixed regressor.

Remark 1 (Scope of a small-gain argument)**.** *Propositions 2 and 3 place the realized command deviation in a known box,* $|\delta_{k,i}| \le \beta\ell_i$*, uniformly in* k *and independently of what the decoder has learned. If the baseline loop is input-to-state stable with respect to an additive torque perturbation, with gain* $\gamma(\cdot)$ *on that input, then the augmented loop inherits an ISS estimate with asymptotic bound* $\gamma(\beta\|\boldsymbol{\ell}\|)$*, and* β *tunes it continuously to zero. Two conditions are not discharged here. The perturbation entering the plant is post-allocation, so the relevant bound is on* B^{-1} *applied to* $\boldsymbol{\delta}_k$ *under rotor saturation, not on* $\boldsymbol{\delta}_k$ *itself; and the required ISS gain for the saturated quaternion-PD baseline is not established in this paper. The architecture therefore supplies the bounded perturbation such an argument needs, and identifies the two missing ingredients, without asserting the conclusion.*

D. Nengo Realization

In Nengo 4.1.0, CMG is a custom non-spiking `NeuronType` with probeable membrane and gate states. Constructed currents are injected into `ensemble.neurons`; conventional NEF gain/intercept fitting is bypassed. The result is a direct-current neural basis with a Nengo PES decoder, not a claim that the full classical controller has been compiled into an NEF population representation.

With unfiltered activity and $n = 2m$, Nengo's update scaling is matched using $\eta_N = \eta n/h$ and Nengo error $\boldsymbol{s}_k - \boldsymbol{y}_k^*$. The implementation subtracts $\lambda_W W_k$ using the pre-step weights and projects after the simulator step. The measured worst relative mean-error difference across the ten missions is 2.013×10^{-6}; an isolated command test gives approximately 1.05×10^{-7}. These are numerical comparisons over tested trajectories, not bitwise identity or independent validation of the control hypothesis.

IV. Experimental Protocol and Cross-Mission Results

A. Models, Parameters, and Measurements

MuJoCo supplies rigid-body and actuator simulation [16]. The X2 model uses four rotor actuators. The CF2 case is an explicit four-rotor design with $1.5\times$ nominal upper thrust authority; it is neither stock-Crazyflie hardware validation nor a validated motor upgrade. For both rotor models, $\boldsymbol{\ell} = \max(\mathrm{diag}(J)\odot(12,12,10)^\mathsf{T}, 10^{-6}\mathbf{1})$ in N m, with the maxima taken componentwise.

TABLE IV
Canonical 3-channel CMG–PES parameters. Neural states/currents are normalized model variables; learning coefficients are per step.

Quantity	Value
h, duration	0.0005 s, 45 s
τ, τ_r	0.05 s, 0.005 s
$v_{\mathrm{rest}}, v_{\mathrm{reset}}, v_{\mathrm{th}}$	$0, 0, 1$
$v_{j,0}, r_{j,0}, \chi$	$0, 0, 1$
$\alpha_h, \gamma, g_0, g_1$	$50, 2, 120, 30$
I_b, I_g, G	$0.75, 1.25, 10^5$
$\beta, \eta, \lambda_W, W_{\max}$	$0.5, 8\times10^{-4}, 10^{-6}, 6$
k_e, k_ω^*	$8, 0$

Each baseline/CMG pair uses the same initial pose/rate, reference, duration, allocator, model, and timestep. The primary campaign is nominal and deterministic, with no injected disturbances or sensor noise. One run per cell is reported; repeated numerical integration of the same deterministic case is not statistical replication. Learning remains active throughout evaluation. We report the sample mean $\bar{\theta}_q$ and relative reduction $100(\bar{\theta}_{q,\mathrm{PD}} - \bar{\theta}_{q,\mathrm{CMG}})/\bar{\theta}_{q,\mathrm{PD}}$. Summary metrics use all 90,000 integration samples; most saved plotting traces retain every tenth sample (5-ms spacing), while the X2 Hard45 trace is full-rate. Thus recomputing a mean from the decimated plot data need not exactly reproduce the full-rate metric.

The five reference triples are generated as follows (radians): Roll is $(0.25\sin 0.7t, 0, 0)$; Full3 is $0.5\sin(t + (0, \pi/2, \pi)^\mathsf{T})$, componentwise. Hard45 is

$$\begin{bmatrix} 0.55\sin(0.75t) + 0.20\sin(2.10t + 0.40) \\ 0.45\sin(0.95t + \pi/3) + 0.18\sin(1.75t) \\ 0.65\sin(0.55t + \pi) + 0.25\sin(1.35t + 0.20) \end{bmatrix}. \tag{29}$$

Hardstep holds zero for one second, then cycles every two seconds through the rows of

$$\begin{bmatrix} .75 & -.45 & .55 \\ -.65 & .55 & -.70 \\ .50 & .50 & .90 \\ -.80 & -.35 & .20 \end{bmatrix}. \tag{30}$$

Yaw360 is $(0, 0, 2\pi(3\xi^2 - 2\xi^3))$ with $\xi = t/45$. The nominal Hover equilibrium is also available, but its essentially zero error does not support a meaningful percentage comparison.

B. Primary Five-Mission Matrix

All ten primary cases improve mean error (Table V). Gains are largest on Yaw360 and Full3 and smallest on Roll; the abrupt Hardstep case is intermediate. The reported bound-occupancy statistic detects post-clipping commands within the runner's tolerance of a limit. It is not a measurement of requested-wrench infeasibility magnitude. Hardstep has small

TABLE V

CANONICAL NENGO CMG–PES VERSUS PD: MEAN QUATERNION-ANGLE ERROR [RAD], 45 S, $n = 1$ PER CELL. S IS THE LOGGED PERCENTAGE OF SAMPLES WITH AT LEAST ONE ROTOR COMMAND AT A BOUND, PD/CMG. CF2 DENOTES THE 1.5× ROTOR-AUTHORITY DESIGN, NOT STOCK HARDWARE.

Model	Mission	PD	CMG–PES	Reduction	S [%]
X2	Roll	.050565	.041785	17.36%	0/0
	Full3	.303275	.131748	56.56%	0/0
	Hard45	.294082	.138007	53.07%	0/0
	Hardstep	.454528	.359355	20.94%	.016/.239
	Yaw360	.075863	.029902	60.58%	0/0
CF2	Roll	.051337	.042439	17.33%	0/0
	Full3	.288383	.126476	56.14%	0/0
	Hard45	.285449	.132319	53.65%	0/0
	Hardstep	.426182	.347886	18.37%	.137/.264
	Yaw360	.075812	.029870	60.60%	0/0

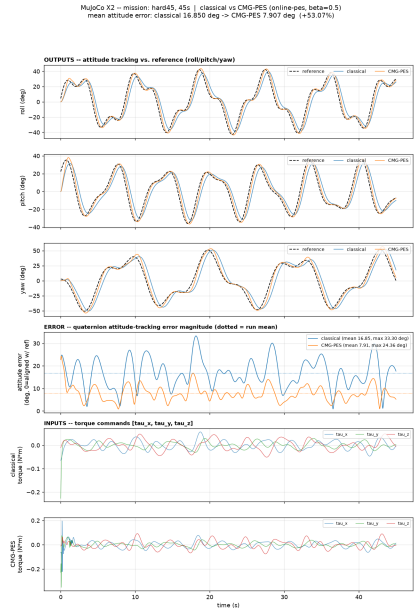


Fig. 4. X2 Hard45: classical baseline vs. CMG–PES residual control.

but nonzero occupancy on both models, so a blanket zero-saturation claim would be incorrect. These cases establish repeatable nominal performance; robustness to model mismatch and comparison against other baseline families are separate questions, addressed in Sec. V. Figures 4 and 5 show the X2 Hard45 case, the largest smooth-mission gain, together with the neural activity that produced it. Hardstep, the smallest reduction and the only case with nonzero bound occupancy, is reported in Tables V and VI.

C. Sensitivity to Measurement Noise

Table V uses noise-free feedback and one run per cell. Every cell was therefore rerun with zero-mean Gaussian noise on the attitude and rate estimates, at two levels and two seeds (Table VI).

Across the twenty noisy cells the seed standard deviation of the reduction is 0.002–0.031%, three orders of magnitude below the effect: under this disturbance model the missions

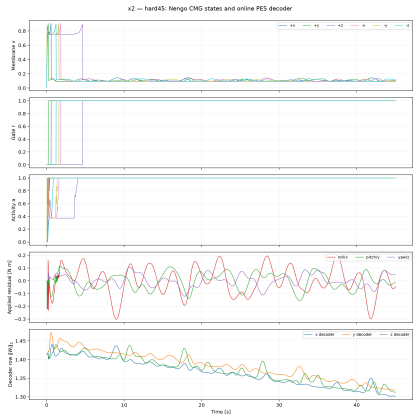


Fig. 5. X2 Hard45: neural activity for the CMG residual controller.

TABLE VI

MEAN-ERROR REDUCTION VERSUS PD UNDER MEASUREMENT NOISE. ENTRIES ARE X2/CF2. CLEAN REPRODUCES TABLE V; MODERATE IS 0.010 RAD AND 0.025 RAD S^{-1}, REALISTIC IS 0.020 RAD AND 0.050 RAD S^{-1} ON THE ATTITUDE AND RATE ESTIMATES. NOISY ENTRIES AVERAGE TWO SEEDS; THE SEED STANDARD DEVIATION IS GIVEN FOR THE REALISTIC LEVEL.

Mission	Clean	Moderate	Realistic
Roll	17.4/17.3%	32.3/32.4%	32.2/32.5% ±0.01
Full3	56.6/56.1%	56.4/55.9%	55.7/55.2% ±0.00
Hard45	53.1/53.6%	52.9/53.4%	52.3/52.7% ±0.01
Hardstep	20.9/18.4%	20.8/18.3%	20.6/18.1% ±0.02
Yaw360	60.6/60.6%	56.8/56.8%	47.8/47.8% ±0.03

are effectively deterministic, and noise *magnitude*, not seed, is the axis that matters. The magnitude response is not uniform. Full3, Hard45, and Hardstep move by under one point. Yaw360 loses 12.8 points on both models: its absolute error is the smallest in the matrix, so a fixed noise floor takes a larger share of the ratio. Roll moves the other way, 17.4% to 32.2%, consistent with the residual having more to correct once the baseline is disturbed. The realistic-noise range is 18.1–55.8%, narrower at both ends than 17.3–60.6%.

D. What Is Learning?

Fixed-decoder residuals already improve some missions: in the audited direct X2 runs their mean reductions range from about 0.12% to 10.27%; the corresponding CF2-design range is 0.06% to 6.80%. Thus it is inaccurate to say that fixing the decoder removes *all* improvement. PES produces the much larger 17.3–60.6% primary reductions. Separately, toggling χ with the decoder mode held fixed is numerically indistinguishable under the tested settings. Decoder adaptation is therefore the principal lever in the tested regime.

Because $a_j = \mathrm{clip}(10^5 r_j, 0, 1)$, decoder activity saturates already at $r_j = 10^{-5}$. Tiny but nonzero gates can therefore yield order-one activity while $c_j \propto r_j^2$ remains small. This is a plausible mechanistic explanation for the observed conductance insensitivity, not a claim that every gate remains $O(10^{-6})$ at every instant. A calibrated gate-gain sweep and gate/membrane readout ablation are needed to determine

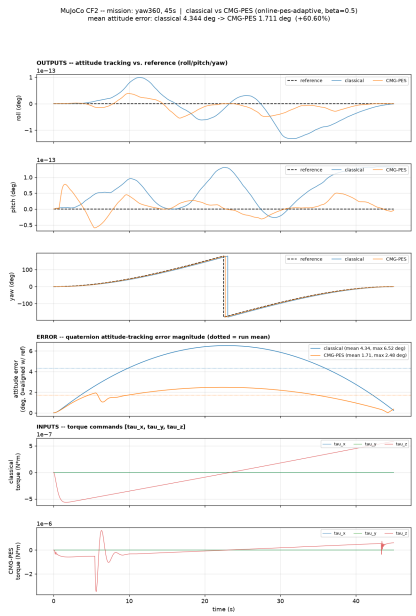


Fig. 6. CF2 Yaw360: classical baseline vs. CMG–PES residual control.

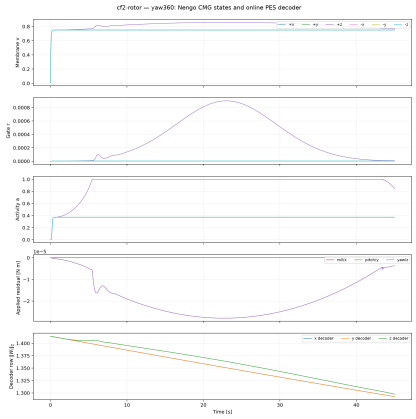


Fig. 7. CF2 Yaw360: neural activity for the CMG residual controller.

whether CMG offers benefits beyond a simpler dynamical basis.

Figure 6 shows the CF2 Yaw360 case, the largest reduction in Table V, and Fig. 7 distinguishes the membrane, raw gate, and clipped activity using that same recorded trajectory rather than equating them.

E. Disturbance Rejection and Retention

A complementary test holds the equilibrium attitude and applies a multi-sine body torque in bursts, so the controller is seen entering a disturbance, adapting inside it, and then holding a decoder adapted to a disturbance that has stopped (Fig. 8).

Before the first burst the residual norm is $O(10^{-16})$: the augmented loop is inert when the baseline is already exact. During bursts the mean error falls from 11.50° to 6.65° (42.2%), and during the quiet intervals from 1.21° to 0.99°

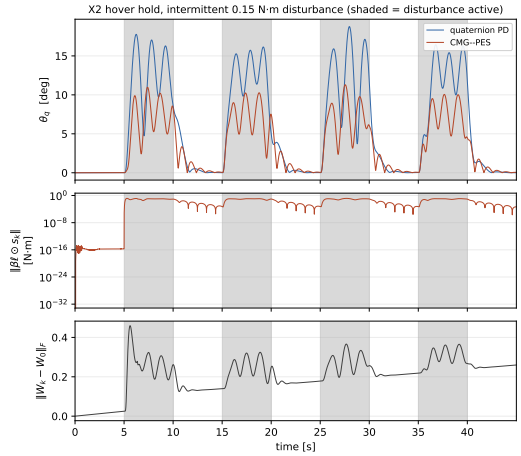


Fig. 8. X2 hover hold under four $0.15\,\mathrm{N\,m}$ disturbance bursts (shaded) separated by quiet intervals. Top: attitude error. Centre: residual magnitude, logarithmic. Bottom: decoder displacement from W_0. The residual is at machine precision before the first burst, so the augmented loop costs nothing while the baseline is already exact.

(18.0%). The same study on CF2, with the disturbance scaled by the inertia ratio, gives 50.2%.

Over $180\,\mathrm{s}$ and sixteen bursts the burst-phase reduction stays flat (43.3% to 40.6%) while the quiet-phase reduction improves from 25.6% to 63.4% and $\|W_k - W_0\|_F$ grows from 0.14 to 0.75. The decoder accumulates a correction reused across episodes rather than relearning each one. Two caveats: in the second episode the augmented loop is 4% *worse* in the quiet phase, the only such interval observed; and the displacement is still growing at $180\,\mathrm{s}$, with $\max_i |W_{k,i}| = 1.09$ inside $\mathcal{W}$ but not saturated, so retention is not shown to converge on longer missions.

F. Neuron-Family Comparison

The Nengo benchmark contains two models, five missions, and eleven controllers: PD and fixed/PES variants of CMG, LIF, ALIF, Izhikevich [22], and LIFRate. All use the same residual/allocator path, with model-specific native activity and filtering. Some Izhikevich–PES cells outperform CMG–PES on mean tracking error. The audit reports an approximate average normalized residual RMS of 0.37 for Izhikevich–PES versus 0.12 for CMG–PES, alongside more decoder movement. Equal output caps do not imply equal realized correction effort. Very sparse LIF/ALIF activity in some tested cells is an operating-point limitation, not evidence that those neuron families cannot learn. The supported conclusion is therefore a tracking–effort tradeoff: CMG–PES attains comparable tracking at markedly lower realized residual effort.

V. Discussion and Conclusion

A fixed quaternion-PD law augmented by six paired continuous membrane–gate units and eighteen online-adapted decoder weights reduces mean attitude error by 17.3–60.6% across ten matched 45-s cases on two modeled platforms, with actuator bounds respected throughout and the baseline

recovered exactly at zero authority. The direct and Nengo realizations agree to 2.013×10^{-6}, so the result is a property of the architecture rather than of one implementation.

The ablation is the more informative half. Holding the decoder mode fixed and toggling conductance modulation is numerically indistinguishable under the tested settings, while freezing the decoder removes most of the gain. In this regime the improvement is carried by online decoder adaptation over a bounded nonlinear basis, not by the specific neuron dynamics. Because $a_j = \mathrm{clip}(10^5 r_j, 0, 1)$, activity saturates once r_j exceeds 10^{-5}, which plausibly explains that insensitivity and identifies the gate gain as the parameter a follow-up should sweep first. Reporting this constrains the architecture's claim rather than supporting it, and it is the reason the contribution is stated as a controlled separation of mechanisms.

Three limits bound the above. The study is in simulation: the noise sweep of Sec. IV-C bounds sensitivity to measurement error and shows the missions are effectively deterministic under it, but it does not address model mismatch, and retention in Sec. IV-E has not converged by $180\,\mathrm{s}$. The comparison is against a well-tuned PD without integral action, not against every reasonable baseline. And bounded commands, states, and weights are implementation properties, not stability proofs; Remark 1 states exactly which two ingredients an ISS argument still needs.

The continuous regime suggests a hardware direction that a spiking formulation forecloses. Membrane leak, sigmoidal gating, and a shunting conductance are the primitives analog neuromorphic circuits implement as device physics [8], with no comparator, reset, or event-routing logic, and the gate transient carries the timing information a spike would otherwise discretize. Whether that translates into an energy advantage is untested and does not follow from these results: a graded analog system draws static rather than event-sparse power, and at six neurons and eighteen weights the controller is negligible on any substrate. The claim such a study would have to make is a per-decision energy comparison at matched tracking accuracy, including conversion and sensing; online decoder adaptation is attractive there mainly because it can absorb device mismatch, which is the standard argument for learning on analog hardware. We flag this as motivation for the continuous formulation, not as a result.

Preliminary real-time tests on a Quanser 3DOF Hover confirm the controller meets a 200-Hz hardware deadline with ten 0.5-ms neural substeps per sample; three matched Roll pairs improved mean error by $2.65\% \pm 1.08\%$. That rig is a different plant from either simulated airframe, so those runs are reported as feasibility rather than validation and are analyzed separately. Follow-up work includes an integral-action baseline, a matched-effort comparison against non-neural bounded bases, a gate-gain sweep, longer disturbance horizons to test whether retention saturates against $\mathcal{W}$, and frozen-weight held-out tests.

REFERENCES

[1] T. Johannink, S. Bahl, A. Nair, J. Luo, A. Kumar, M. Loskyll, J. A. Ojea, E. Solowjow, and S. Levine, "Residual reinforcement learning for robot control," in *2019 International Conference on Robotics and Automation (ICRA)*, 2019, pp. 6023–6029.

[2] Z. Wu, S. Cheng, P. Zhao, A. Gahlawat, K. A. Ackerman, A. Lakshmanan, C. Yang, J. Yu, and N. Hovakimyan, "L1Quad: L_1 adaptive augmentation of geometric control for agile quadrotors with performance guarantees," *IEEE Transactions on Control Systems Technology*, vol. 33, no. 2, pp. 597–612, 2025.

[3] X. Li, B.-C. Zheng, W. Wang, and W. Zhang, "HPA-PPO: A history-preview and attention-enhanced PPO for quadrotor attitude tracking," *Applied Mathematics and Computation*, vol. 532, p. 130236, 2027.

[4] T. Bekolay, J. Bergstra, E. Hunsberger, T. DeWolf, T. C. Stewart, D. Rasmussen, X. Choo, A. R. Voelker, and C. Eliasmith, "Nengo: A Python tool for building large-scale functional brain models," *Frontiers in Neuroinformatics*, vol. 7, p. 48, 2014.

[5] T. DeWolf, T. C. Stewart, J.-J. Slotine, and C. Eliasmith, "A spiking neural model of adaptive arm control," *Proceedings of the Royal Society B: Biological Sciences*, vol. 283, no. 1843, p. 20162134, 2016.

[6] T. DeWolf, P. Jaworski, and C. Eliasmith, "Nengo and low-power AI hardware for robust, embedded neurorobotics," *Frontiers in Neurorobotics*, vol. 14, p. 568359, 2020.

[7] A. Olivares, E. S. Espinoza, L. E. Ramos, O. A. G. A., L. R. G. Carrillo, and A. T. Sornborger, "Self-tuning neuromorphic controller for real-time UAS trajectory tracking based on prescribed error sensitivity," *Neural Computing and Applications*, vol. 38, no. 7, p. 192, 2026.

[8] M. Mahowald and R. Douglas, "A silicon neuron," *Nature*, vol. 354, pp. 515–518, 1991.

[9] A. R. Voelker, "A solution to the dynamics of the prescribed error sensitivity learning rule," Centre for Theoretical Neuroscience, University of Waterloo, Tech. Rep. CTN-TR-20151001-01, Oct. 2015. [Online]. Available: https://www.compneuro.uwaterloo.ca/files/publications/voelker.2015.pdf

[10] Google DeepMind, "MuJoCo Menagerie," https://github.com/google-deepmind/mujoco_menagerie, model repository; accessed September 5, 2026.

[11] ——, "Bitcraze Crazyflie 2 Description (MJCF)," https://github.com/google-deepmind/mujoco_menagerie/blob/main/bitcraze_crazyflie_2/README.md, model provenance and actuator-limit caveat; accessed September 5, 2026.

[12] Bitcraze AB, "PWM to Thrust: Old Measurements (2015)," https://www.bitcraze.io/documentation/repository/crazyflie-firmware/master/functional-areas/pwm-to-thrust/, historical test setup and total-thrust table; accessed September 5, 2026.

[13] ——, "Crazyflie 2.0: Mechanical and Flight Specifications," https://store.bitcraze.io/products/crazyflie-2-0, accessed September 5, 2026.

[14] Skydio, "Skydio X2E Datasheet," https://pages.skydio.com/rs/784-TUF-591/images/skydio-x2e-datasheet-x2-pg.pdf, aircraft specifications; accessed September 5, 2026.

[15] Bitcraze AB, "Measuring Propeller RPM: Part 3," https://www.bitcraze.io/2015/02/measuring-propeller-rpm-part-3/, Feb. 2015, published February 9, 2015; accessed September 5, 2026.

[16] E. Todorov, T. Erez, and Y. Tassa, "MuJoCo: A physics engine for model-based control," in *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2012, pp. 5026–5033.

[17] MuJoCo Developers, "MuJoCo Fluid Forces: Inertia Model," https://mujoco.readthedocs.io/en/3.1.1/computation/fluid.html, accessed September 5, 2026.

[18] C. G. Mayhew, R. G. Sanfelice, and A. R. Teel, "Quaternion-based hybrid control for robust global attitude tracking," *IEEE Transactions on Automatic Control*, vol. 56, no. 11, pp. 2555–2566, 2011.

[19] MuJoCo Developers, "MuJoCo XML Reference: Body/Inertial," https://mujoco.readthedocs.io/en/stable/XMLreference.html#body-inertial, accessed September 5, 2026.

[20] E. Fresk and G. Nikolakopoulos, "Full quaternion based attitude control for a quadrotor," in *2013 European Control Conference (ECC)*, 2013, pp. 3864–3869.

[21] J. C. Monteiro, F. Lizarralde, and L. Hsu, "Optimal control allocation of quadrotor UAVs subject to actuator constraints," in *2016 American Control Conference (ACC)*, 2016, pp. 500–505.

[22] E. M. Izhikevich, "Simple model of spiking neurons," *IEEE Transactions on Neural Networks*, vol. 14, no. 6, pp. 1569–1572, 2003.